

■ MongoDB Atlas & Mongoose Connection Guide

A clean step-by-step summary reference guide from today's class process.

■■ Step 1: Set Up Cloud Database (MongoDB Atlas)

- **Create Account:** Register and log in to your account dashboard on MongoDB Atlas.
- **Deploy Cluster:** Deploy a free shared database tier cluster using default name **Cluster0**.
- **Configure Credentials:** Set up database user access configuration metrics:
 - **Username:** admin
 - **Password:** admin@123

■■ Step 2: Structure the Remote Database

- **Database Creation:** Inside the Data Explorer interface, build a logical database explicitly named **Database1**.
- **Collection Initialization:** Initialize an empty data collection named **User** inside that database.

■ Step 3: Local Environment Installation & Setup

- **Install Package:** Open the terminal within your NodeServer project directory path and install the package framework instance dependency via node package manager:

```
npm install mongoose
```

- **Import Module:** Required reference context header invocation statement inside application main scripts (e.g., index.js):

```
const mongoose = require('mongoose');
```

■ Step 4: Establish Connection with URL Encoding

Because the database password value contains a reserved string operator delimiter token feature character (@), you must escape parsing layers by replacing it with its hexadecimal percent escape value **%40**. Failure to handle this will return a *MongoParseError: Invalid connection string* crash dump trace.

```
mongoose.connect('mongodb+srv://admin:admin%40123@cluster0.p1jxd0c.mongodb.net/Datab  
ase1?retryWrites=true&w=majority&appName=Cluster0')  
  .then(() => {  
    console.log('Connected to MongoDB successfully!');  
  })  
  .catch((err) => {  
    console.error('Error connecting to MongoDB:', err);  
  });
```